AI CLUSTERS

Next Generation Optically Integrated Platform Solution for AI Data Centers

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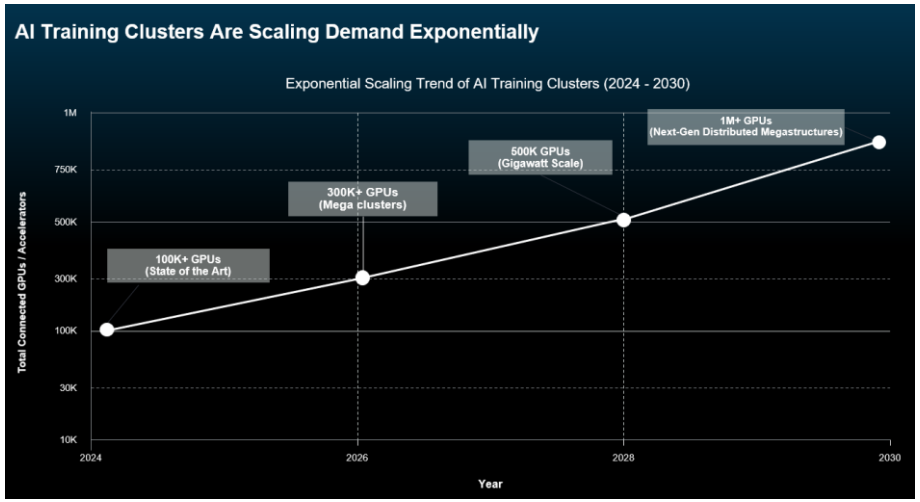
OCP PLATINUM Member

Agenda

1. AI Cluster Scaling challenges
2. Co-Packaged Optics (CPO) enablement
3. OPEN Optically Integrated Platform Solutions

AI Cluster Scaling Demand

- Driven by the race to train frontier models and deploy massive inferencing workloads
- Expanding clusters to 500K/1M+ accelerators
- Shifting engineering focus from simple chip counts to entire multi-rack-scale, data center solutions
  - Power and cooling
  - Interconnect and fabric latency
  - Physical reach limits of cabling infrastructure





AI Fabric Scaling Challenges

- Fabric challenges are not resolved in bandwidth alone, but need predictable performance, visibility and serviceability
- Scale-Out and Scale-Up have different architectural and operational requirements
- Traditional approaches face increasing density and power constraints
  - AI fabric are expanding across rack and inter-rack boundaries

TODAY

Scale-Up
Copper Interconnect
DAC / AEC



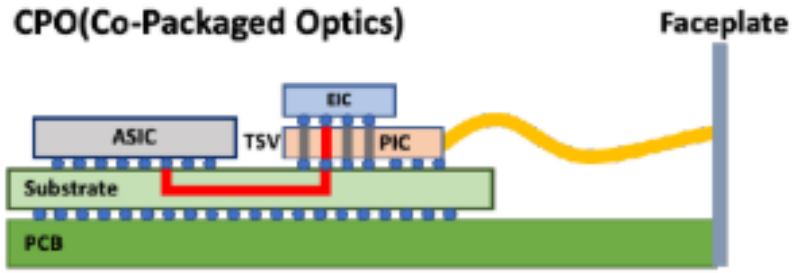
Scale-Out
Pluggable
Optics



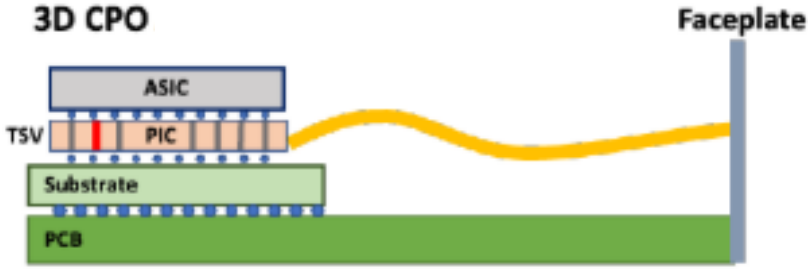
Co-Packaged Optics (CPO) Enablement

- Technically advantageous integration → pj/bit
- Quality and reliability concerns are being disproved
- Support for both Scale-Out and Scale-Up
- Ecosystem solutions are becoming readily available

Gen 1 CPO



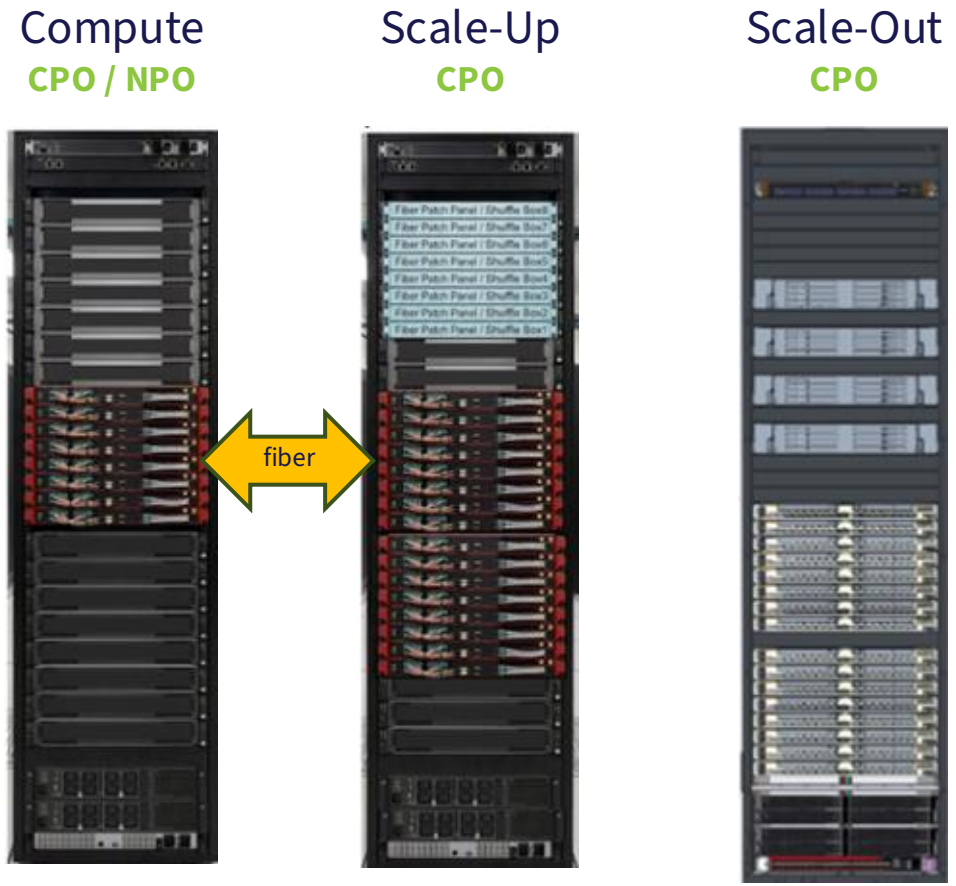
CPO Evolution



CPO for AI Clusters

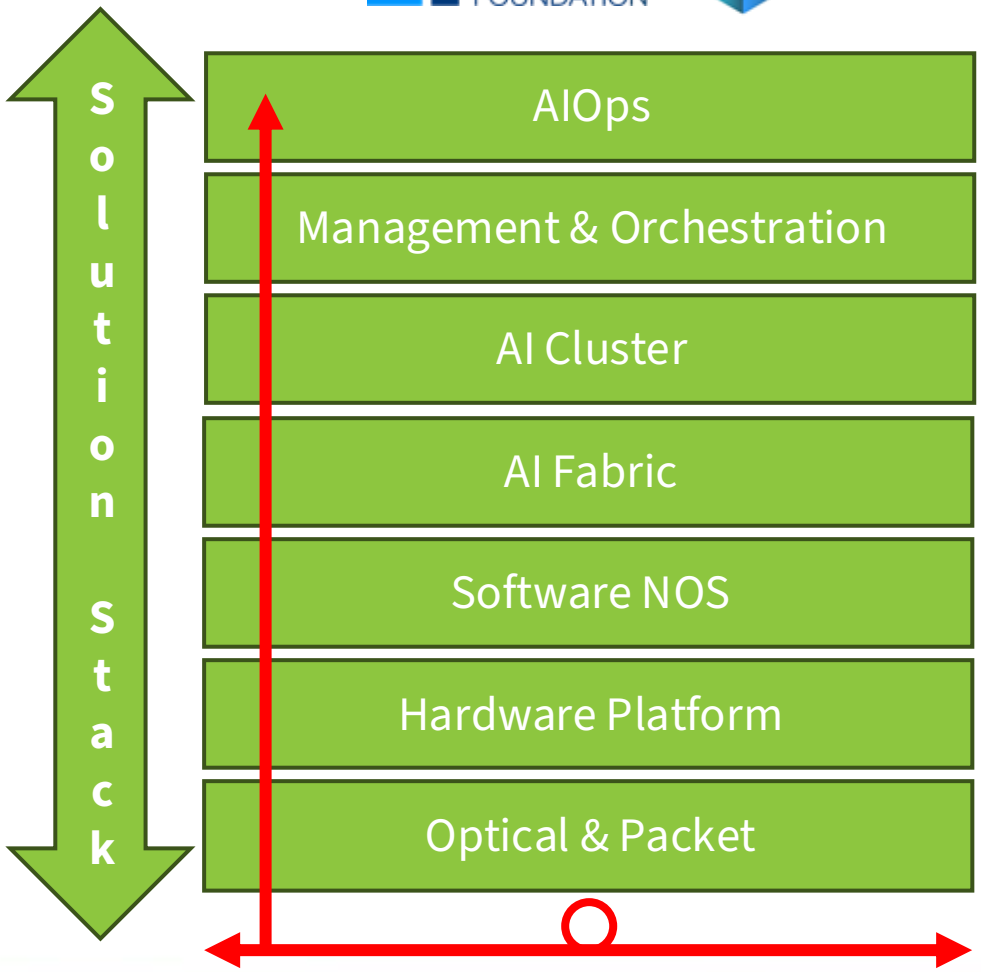
- CPO is an enabling technology, but not the complete solution
- New requirements for thermal management, monitoring, serviceability, qualification and fault isolation
- **Value comes from integrating optics, networking software, standards, telemetry and operations**

TOMORROW



Enabling OPEN Platform Solutions

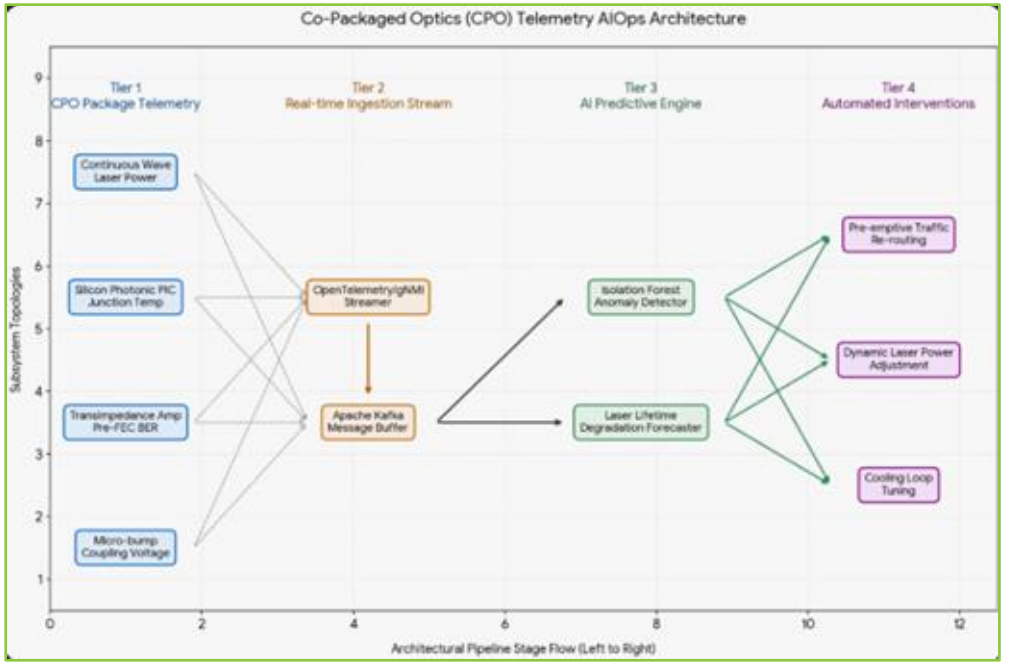
- OCP Open Cluster Designs for AI
- OCP Open Silicon Photonics for AI Systems
- Linux Foundation SONiC
- OCP Switch Abstraction Layer (SAI)
- OCP Switch Platform Architecture
  - A Unified Network Switch Platform Reference Design
- OCP ESUN and Ultra Ethernet Consortium (UEC)
  - Ethernet Scale-Up and Scale-Out
- OCP Enhanced Telemetry Optics Product Software
- OIF CMIS for CPO and ELSFP module



Solving the Telemetry Challenges

Common Operational Model to Reduce Complexity Across AI Fabric domains:

- Reuse consistent telemetry concepts across Scale-Out and Scale-Up
- Provide common visibility across optical and packet layers
- Define clear ownership between platform, network operations, and hardware teams
- Support repeatable deployment, qualification, troubleshooting, and lifecycle processes



Critical Metrics for CPO Scale-Out and Scale-Up Fabrics
• Thermal
• Optical power
• External laser performance
• Pre-FEC and Post-FEC BER

Celestica’s DS6005 OPEN CPO Platform Solution

- AI Fabric solution based on Broadcom’s Tomahawk 6 Davisson CPO chipset



OPEN Compute Project SOLUTION PROVIDER®

Celestica™

SONiC

- Celestica plans to contribute the design to the OCP Community
- For more information: www.celestica.com/DS6005
- See the DS6005: Booth C2

Call to Action

- Enable OPEN Co-Packaged Optics (CPO) platform solutions
  - **Let's collaborate in accelerating optically integrated AI Fabric solutions**
- Integrate software and telemetry to make CPO operationally useful with its solid architectural foundation
- Collaborate in SONiC, UEC, ESUN and OCP forums to help the ecosystem advance interoperable AI-fabric deployments
- Potential OCP discussion areas:
  - **Common telemetry models and data definitions**
  - **Interoperable APIs and abstractions**
  - **Consistent health and fault semantics**
  - **Qualification and validation methods**
  - **Failure-domain visibility**
  - **Serviceability and lifecycle workflows**
  - **Multi-vendor interoperability**

Thank You!

